BUS 792: Applied Business Analytics

Assignment (1)

Preferably using Python

Part 1 (Data Visualization)

Choropleth map (a map that uses differences in shading, coloring, or the placing of symbols within predefined areas to indicate the average values of a property or quantity in those areas) is an interesting data visualization tool.

Data source for this problem can be found here:
https://raw.githubusercontent.com/TrainingByPackt/Interactive-Data-Visualization-with-Python/master/datasets/share-of-individuals-using-the-internet.csv

You will use the **express** module from **plotly** and use the **choropleth** function from the module.

(0) download the data from its source.
(1) Since the DataFrame contains records from multiple years, Create a subset of the data to one specific year, say 2016, using the .query().
(2) Generate an interactive worldwide choropleth map using the subset data from part (1) and the **choropleth** function of plotly library.
The following parameters to be set:

- locations: This is set to the name of the column in the DataFrame that contains the country codes, that is, Code
- color: This is set to the name of the column that contains the numerical feature using which the map is to be color-coded, that is, (Individuals using the Internet (% of population)

- hover_name: This is set to the name of the column that contains the feature to be displayed while hovering over the map, that is, Country

- color_continuous_scale: This is set to a color scheme, that is, px.colors.sequential.Plasma.

(3) What can you say about the internet usage in the western world compared to the east?
(4) How the percentage of individual using internet in Canada and Australia is compared to most European countries?
(5) Add title text to the choropleth map setting the title_text parameter, that is, fig.update_layout(title_text='Internet usage across the world (% population) -2016')
(6) Suppose we are only interested in seeing internet usage across the continent of Asia, we can do this setting geo_scope parameter of the fig.update_layout to any of the contitnet, e.g. north america | south america | africa | asia | europe | usa.
(7) Add title text to the choropleth map setting the title_text parameter, that is, fig.update_layout(title_text='Internet usage across the world (% population) -2016')
(8) Let the plot **rotate** like a real globe. For that, Set the projection type to natural earth, that is, geo = dict(projection={'type':'natural earth'}) '
(9) How internet usage has grown in different countries of the world over the years? We could repeat the above for every year. However, we can use the paramter animation_frame, that is, animation_frame="Year", to the name of the column whose values we wish to animate our visualization for. You have to change the data frame to use the whole dataset not the subset created in (1). Also, since The years on the slider are not in the right order, you have to sort the DataFrame by time (the **Year** feature), that is, internet_usage_df.sort_values(by=["Year"],inplace=True). Comment on the change of the internet usage of years.

Part 2 (Time Series Analysis)

The file **Part_2_TSA.xlsx** contains the weekly total number of loans application in a local branch of a national bank for the last two years. It is suspected that there should be some relationship (i.e., autocorrelation) between the number of applications in the current week and the number of loan applications in the previous weeks. Modelling that relationship will help the management to proactively plan for the coming weeks through reliable forecasts.

Using **Python** and appropriate **packages**:

(a) Plot the time series. By visual inspection, does the data seem to be correlated?

(b) Plot the Autocorrelation function (ACF) and Partial autocorrelation function (PACF) plots.

(c) Based on the plots in (b), suggest an appropriate model to fit.

(d) Report the parameters and their associated confidence intervals (and the P-values).

(e) What is the MSE?

(f) In one figure and 4 subplots, show the 1. Normal Probability plots, 2. Residual versus Fitted value plot, 3. Histogram of the residuals and 4. Time series plot of the residuals.